A Didactic Alignment (N = 4 Taxa, L = 16 Sites) and Branch Code Anatomy

Raw Uncompressed Cost: 4 × 16 × ln 4 = 88.72 nats | Ancestral Root X_r Cost: 16 × ln 4 = 22.18 nats

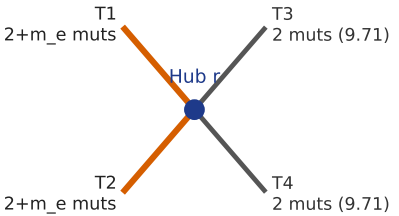
	Shared (1–3)			Private Substitutions (4–11; m_c=2)								Invariant (12–16)					
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Root X_r	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	22.18 nats
Taxon T1	G	G	G	C	C	.	.	.	.	.	.	.	.	.	.	.	m_e + 2 muts
Taxon T2	G	G	G	.	.	T	T	.	.	.	.	.	.	.	.	.	m_e + 2 muts
Taxon T3	.	.	.	.	.	.	.	C	C	.	.	.	.	.	.	.	2 priv. muts
Taxon T4	.	.	.	.	.	.	.	.	.	T	T	.	.	.	.	.	2 priv. muts

Single-Branch Code Length (Eq. 1): $L_{branch}(m, 16) = L_N(m) + \ln C(16, m) + m \ln 3$ $(L_N(0) = \ln 2)$

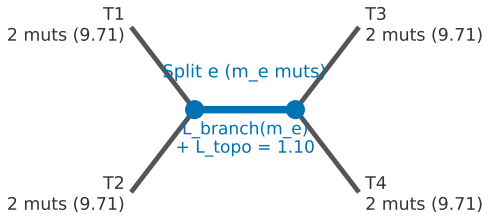
- m = 1 mut: 1.69 + ln(16) [2.77] + 1.10 = 5.56 nats (Unamortized new-branch startup)
- m = 2 muts: 2.73 + ln(120) [4.79] + 2.20 = 9.71 nats (3rd mut on leaf: +3.27 nats)
- m = 3 muts: 3.36 + ln(560) [6.33] + 3.30 = 12.99 nats (m = 4: 15.72; m = 5: 18.05 nats)

B Star Polytomy (T_star) vs. Resolved Quartet (T_split)

Factoring m_e shared A→G sites onto stem e saves $\Delta S(m_e) = 2\,[L_{branch}(2+m_e, 16) - L_{branch}(2, 16)]$



Unresolved Star T_star
(0 internal splits; m_e duplicated in parallel on both T1 and T2)

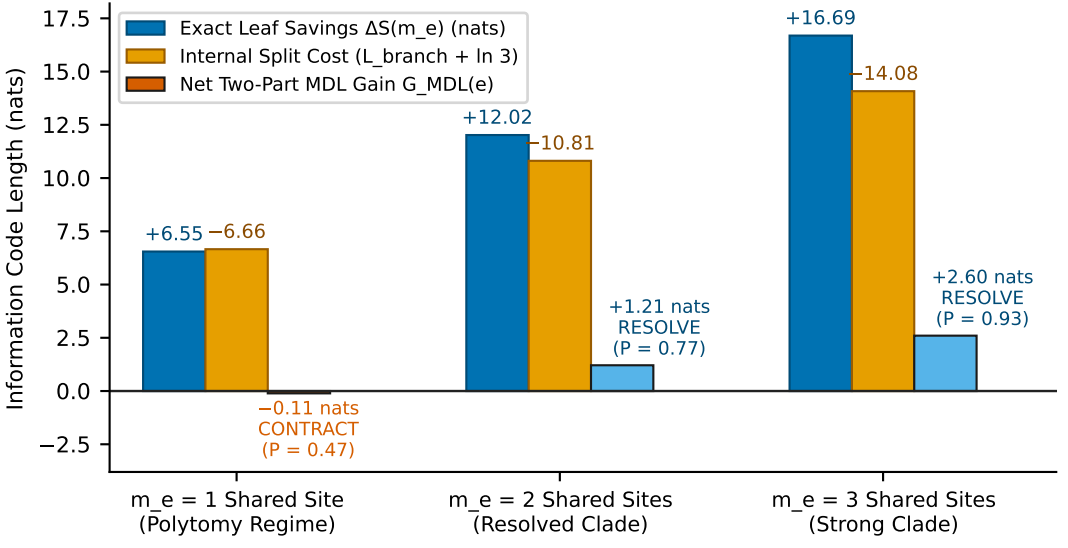


Resolved Quartet T_split
(1 of 3 quartets: L_topo = ln 3 = 1.10; shared muts encoded once on stem e)

Two-Part MDL Split Law (Eq. 5): $G_{MDL}(e) = L_{MDL}(T_{star}) - L_{MDL}(T_{split}) = \Delta S(m_e) - L_{branch}(m_e, 16) - \ln 3$

Split e is retained iff $G_{MDL}(e) > 0$ nats (Posterior Support $P(e \mid A) = 1 / [1 + \exp(-G_{MDL}(e))]$)

C MDL Split Threshold: Why m_e = 1 Contracts while m_e ≥ 2 Resolves



D Asymmetric Branch Weights (LBA Immunity) & Lateral Shunts

1. Why Combinadic Marginal Weights Resist Long-Branch Attraction (Eq. 4)

Suppose T1 & T3 are LONG branches (m_long = 6 muts on L = 16) and T2 & T4 are SHORT (m_short = 1 mut):

- Short-branch positional weight: $w_{short} = \ln[(16-1)/(1+1)] = \ln(7.5) = 2.01$ nats/site
- Long-branch positional weight: $w_{long} = \ln[(16-6)/(6+1)] = \ln(10/7) = 0.36$ nats/site

→ Grouping the two LONG branches {T1, T3} on a chance collision saves only 2 w_long = 0.71 nats of site address, whereas grouping the SHORT branches saves 2 w_short = 4.03 nats (5.6× stronger signal).

2. When Non-Sister Lineages Share Parallel States: Lateral Shunt Peeling (Eq. 7)

Once backbone tree ((T1,T2),(T3,T4)) is fixed, suppose non-sister clades share m parallel replacements:

- Endpoint Address Header: 2 ln |V| nats to specify donor and recipient nodes
- Contiguous HGT Tract: $L_{mask}^{INTERVAL} = \ln 2 + 2 \ln L + L_{branch}(d_{[a,b]}, b-a+1)$
- Scattered Convergence: $L_{mask}^{SPARSE} = \ln 2 + \ln C(M_{pool}, m)$ (certified iff savings > header + mask)